AAR - Layer 5: Response

Confidence-gated, hysteresis-based graduated response (alert -> isolate -> kill)

Akshat Kapur (24BDS0318) · Aahav Yadav (24BKT0138) · Rishab Jajodia (24BDS0268)

github.com/AkshatKapur1/AAR-System-Call-Anomaly-Detection · October 2026

Summary

Layer 5 turns the best model's anomaly scores (LSTM, stream mode) into actions on a running process. Every test recording is replayed window by window as if it were a live process, and actions are taken while it runs. Thresholds are calibrated on normal validation data, settings are chosen on TUNE, and all results are on FINAL (583 normal and 716 attack processes).

1. Results: what happened to each process

Action	Attack processes (of 716)	Normal processes (of 583)
Alerted	**604 (84.4%)**	244 (41.9%)
Isolated (reversible) at some point	102 (14.2%)	13 (2.2%)
Contained (isolated or killed) at some point	**164 (22.9%)**	16 (2.7%)
Still contained at the end	142 (19.8%)	9 (1.5%)
Killed (permanent)	**123 (17.2%)**	**8 (1.4%)**
Left untouched (no isolation or kill)	552	**567 (97.3%)**
System calls before the first action (median)	22 (containment), 28 (kill)	-

1.1 What happened to isolated processes

	Released automatically	Still isolated at the end	Later killed
Normal (13 isolated)	7	1	5
Attacks (102 isolated)	22	19	61

1.2 By dataset

Dataset	Attacks alerted	Attacks contained	Attacks killed	Normal isolated	Normal killed
ADFA-LD (518 attacks / 328 normal)	409	53	32	7	4
LID-DS (81 attacks / 127 normal)	81	61	61	4	3
PLAID (117 attacks / 128 normal)	114	50	30	2	1
CVE-2012-2122 (4 / 62)	4	3	3	4	3
CVE-2014-0160 (77 / 65)	77	58	58	0	0

The response is deliberately graded: the alert line is set for balance, so most attacks are alerted; isolation and kill lines are calibrated so that only ~2% and ~1% of normal processes would ever reach them, and confidence must stay above them for 10 windows. Kills are therefore rare but accurate (123 attacks vs 8 normal processes), and 567 of 583 normal processes are never isolated or killed.

2. How it works

- **Window score**: LSTM stream mode - each new 7-syscall window scored given everything the process did before.
- **Normalise** per dataset with normal val windows: z = (score - median) / (p99 - median).

- **Confidence** = running mean of z since the process started (online form of Layer 4's mean aggregation).
- **Alert** (hysteresis): on above the alert line, off only below it minus 0.25 - stops flicker.
- **Isolate** (reversible): confidence above the isolation line for 10 consecutive windows; released below it minus 0.25. The process keeps running in containment and stays monitored.
- **Kill** (final): confidence above the kill line for 10 consecutive windows - SIGKILL / taskkill.
- A single odd window can never isolate or kill a process (confidence gate).

Parameter	How it is set	Value
Confidence type, sustain	42 settings (running mean / EWMA x sustain 1-100) on TUNE: most attacks killed with <= 2% normal killed	running mean, 10 windows
Kill line per dataset	~1% of normal val processes would ever be killed	ADFA-LD 1.16, LID-DS 0.95, PLAID 1.12
Isolate line per dataset	~2% of normal val processes would ever be isolated	ADFA-LD 1.01, LID-DS 0.88, PLAID 0.82
Alert line	max (% attacks alerted + % normal not alerted) on TUNE	on 0.338, off 0.088

3. Further results (FINAL)

- **Isolation adds containment without adding kills**: 164 attacks contained vs 123 killed, while normal processes killed stay at 8.
- **Hysteresis**: alert raises on normal processes 253 with vs 345 without; 869 vs 1187 over all processes.
- **Live demo**: a real dummy process bound to the switch while replaying Hydra_FTP attack UAD-Hydra-FTP-1-3523 was terminated at window 9 (16 system calls).
- **Check**: simulation and the KillSwitch class agree on 300 / 300 FINAL processes (kill and isolation decisions).

	Layer 5 (online)	Layer 4 strict (whole recording)	Layer 4 balanced (whole recording)
Attacks acted on	123 killed, 164 contained	107	638
Normal processes affected	8 killed, 16 disrupted	6	202
When	while running (median 22-28 syscalls)	after the whole recording	after the whole recording

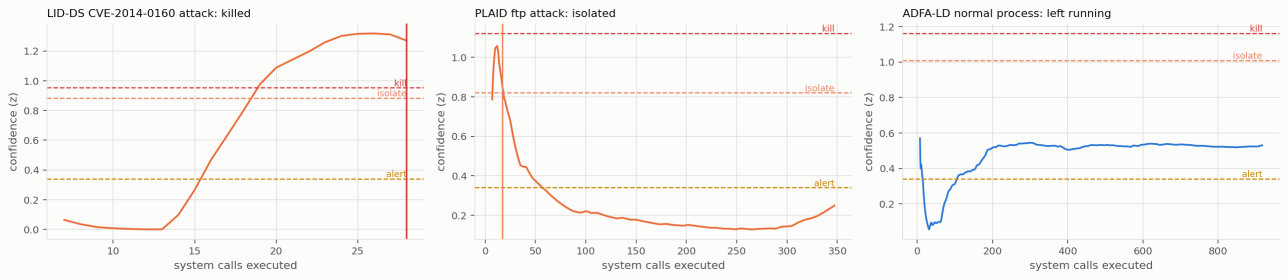


Figure 1 - Confidence over time: an attack killed, an attack isolated, a normal process left running.

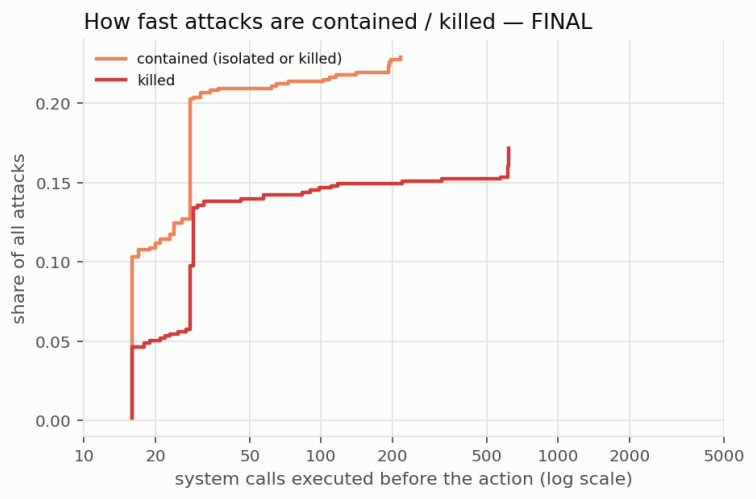


Figure 2 - Share of all attacks contained / killed vs system calls executed.

3.1 Per attack type

Dataset	Attack type	n	Alerted	Contained	Killed	Median syscalls to containment
ADFA-LD	Adduser	64	43	0	0	-
ADFA-LD	Hydra_FTP	113	95	18	16	16
ADFA-LD	Hydra_SSH	122	104	14	4	16
ADFA-LD	Java_Meterpreter	85	65	10	6	16
ADFA-LD	Meterpreter	51	37	1	0	16
ADFA-LD	Web_Shell	83	65	10	6	18
LID-DS	CVE-2012-2122	4	4	3	3	116
LID-DS	CVE-2014-0160	77	77	58	58	28
PLAID	cowroot	11	9	5	0	24
PLAID	ftp	13	13	6	0	192
PLAID	nginx	19	19	17	17	16
PLAID	privesc	4	4	1	1	65
PLAID	redis	48	48	19	12	16
PLAID	ssh	22	21	2	0	193

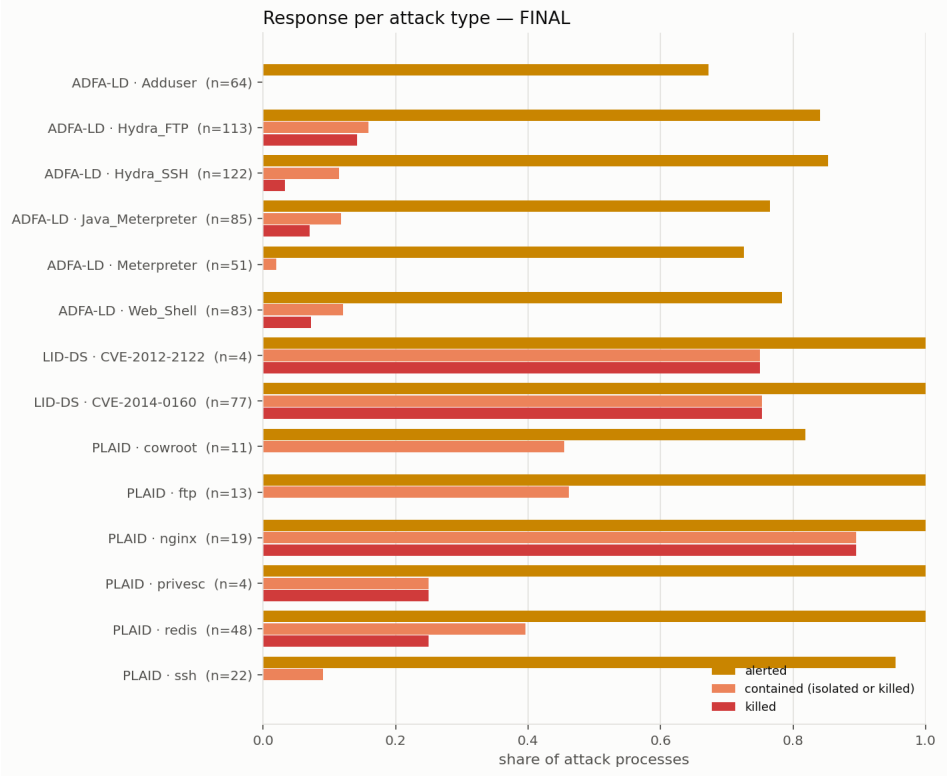


Figure 3 - Alerted, contained and killed per attack type.

4. Options evaluated (TUNE only)

Option	Idea	TUNE outcome	Kept
EWMA confidence	react to recent windows only	fewer kills than the running mean at the same limit	no
Alert escalation	kill when an alert lasts very long	every level broke the 2% kill limit (mildest: +2 attacks for +2 normal kills)	no - removed
Isolate tier	reversible containment between alert and kill	72 vs 46 of 306 attacks stopped, normal kills unchanged (5 of 251)	**yes**

Isolation level (% of normal val)	TUNE attacks stopped	TUNE normal disrupted	TUNE normal killed	Used
2%	72 / 306	14 / 251 (5.6%)	5	yes
3%	97 / 306	20 / 251 (8.0%)	5	
4%	110 / 306	20 / 251 (8.0%)	5	
5%	118 / 306	21 / 251 (8.4%)	5	
7%	134 / 306	26 / 251 (10.4%)	5	
10%	145 / 306	29 / 251 (11.6%)	5	

Disclosure: the pre-set target of at most 5% of normal processes disrupted on TUNE was not met by any isolation level - the most conservative gives 5.6% (14 of 251) - so that level is used and reported as is. A first version of Layer 5 using only EWMA confidence killed 62 attacks with 14 normal processes killed on FINAL; the running-mean option was added afterwards, motivated by Layer 4's choice of mean aggregation, and selected on TUNE.

5. Outputs on GitHub (response/)

File	Contents
calibration.json	normalisation, alert / isolate / kill lines per dataset, confidence type, sustain
tuning_results.csv	42 confidence / sustain settings on TUNE
isolation_sensitivity.csv	isolation levels on TUNE
action_log.csv	one row per FINAL process: alerted, isolated, killed, contained, timings
per_dataset.csv / per_attack_type.csv	breakdowns
response_summary.json	totals, hysteresis effect, Layer 4 comparison, live demo
plots/	example_timelines, time_to_action, per_attack_response

Limitations: simulated replay of recorded traces - live use needs a per-process syscall stream (ptrace/eBPF) and a real sandbox behind isolation; attacks that look normal at the syscall-name level (ADFA-LD Adduser / Meterpreter, PLAID ssh) are alerted but rarely contained; ~1.4% of normal processes are killed, so critical services would need a whitelist or operator confirmation.